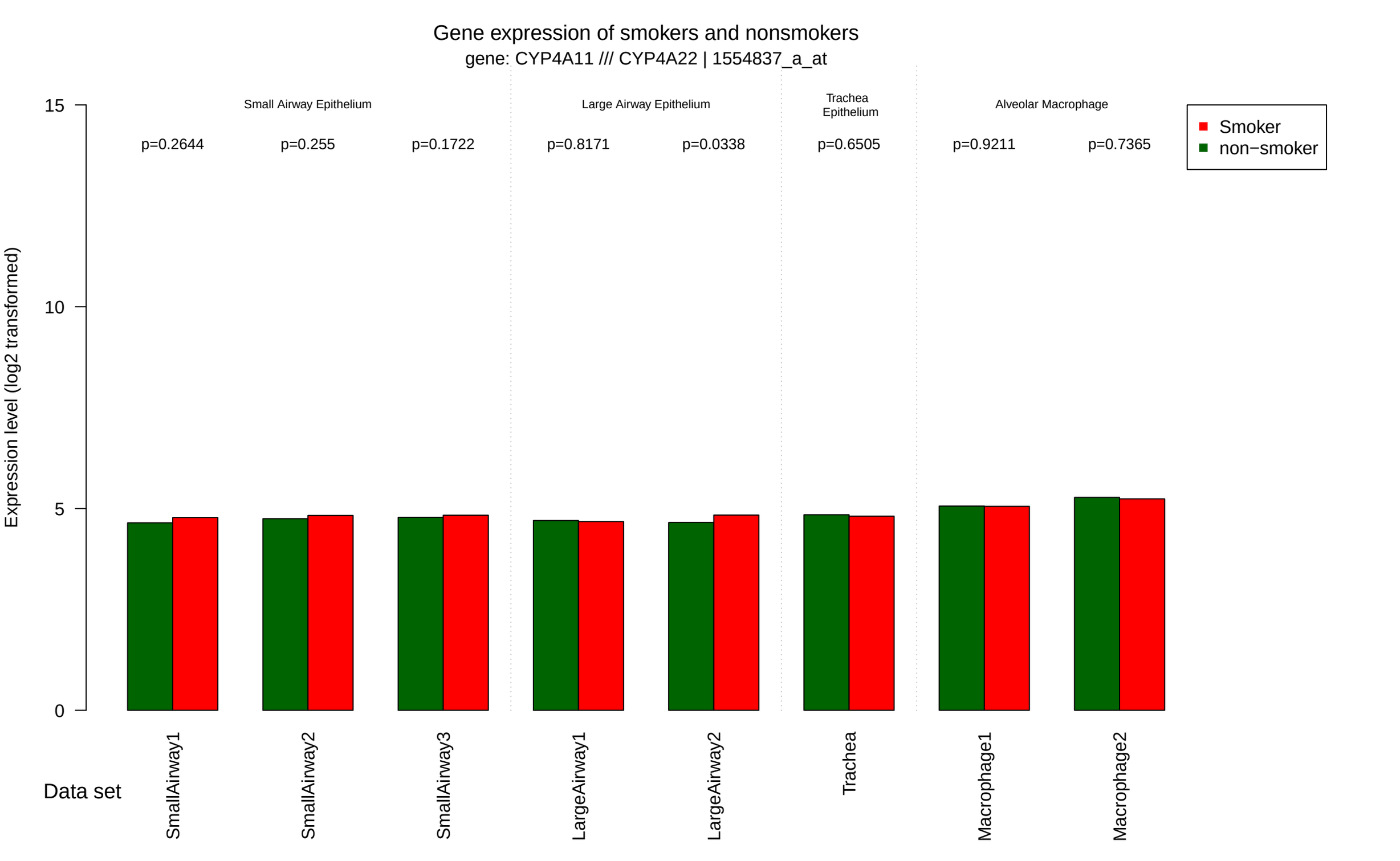
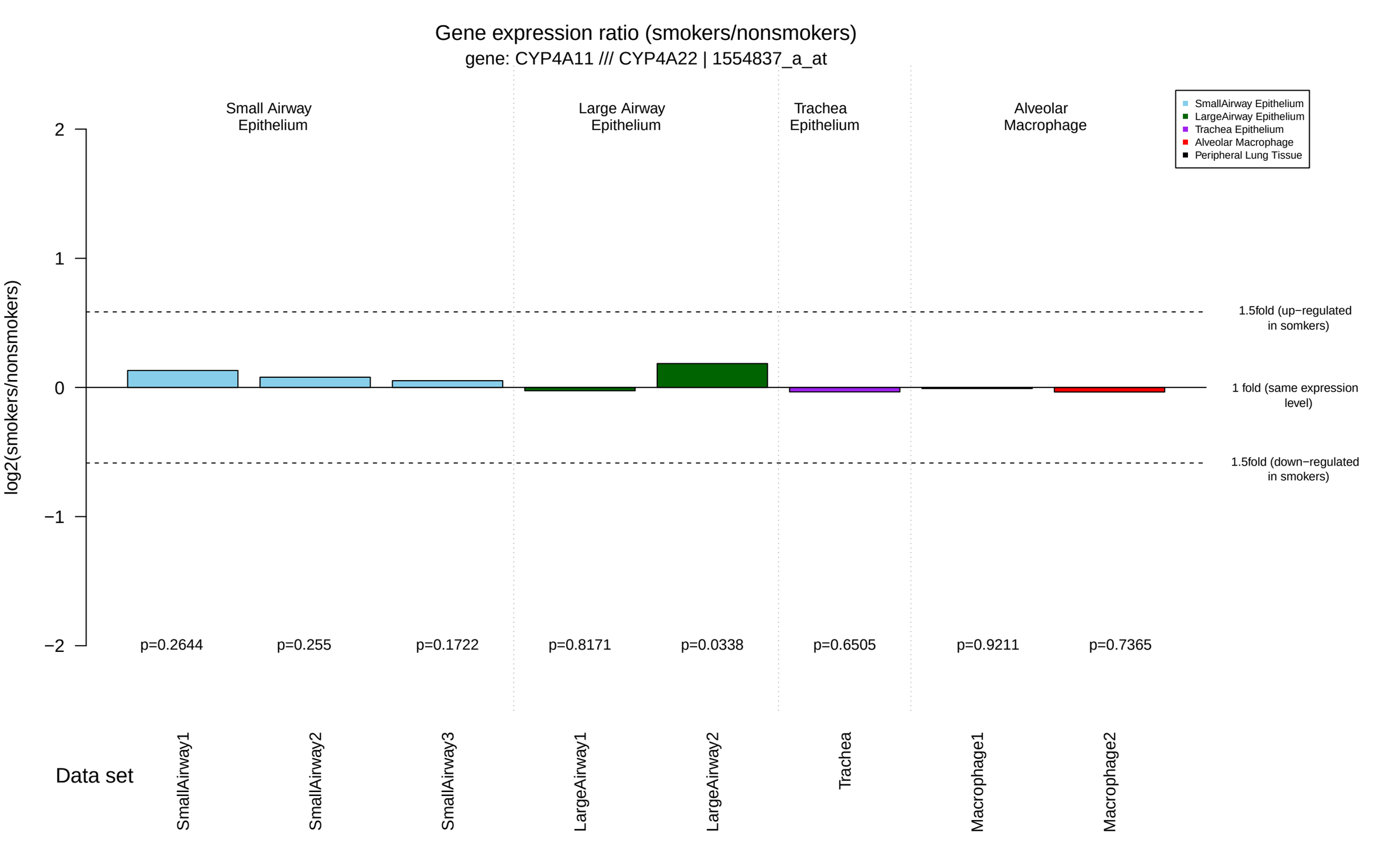
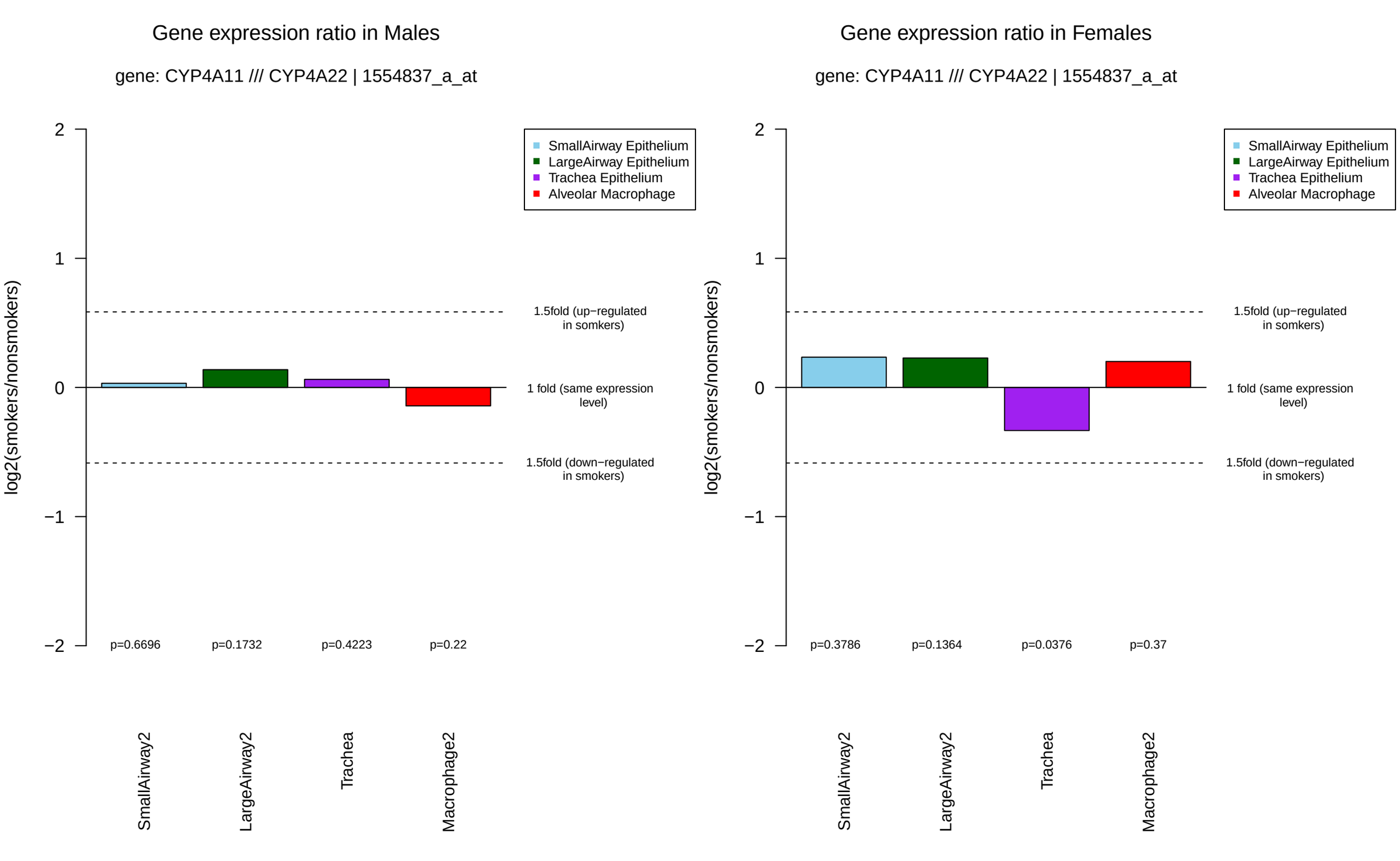
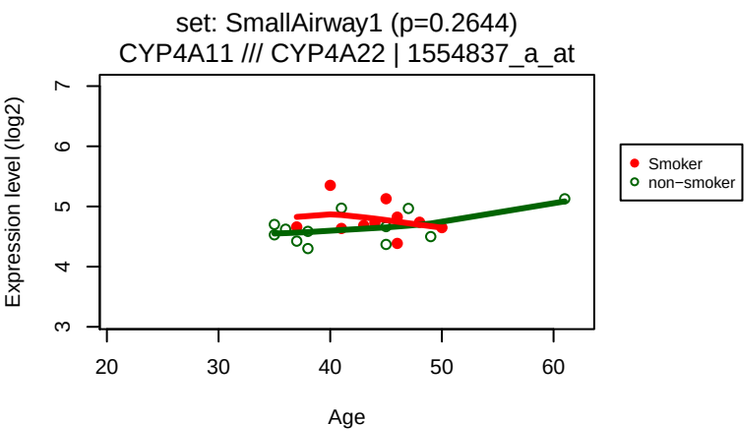
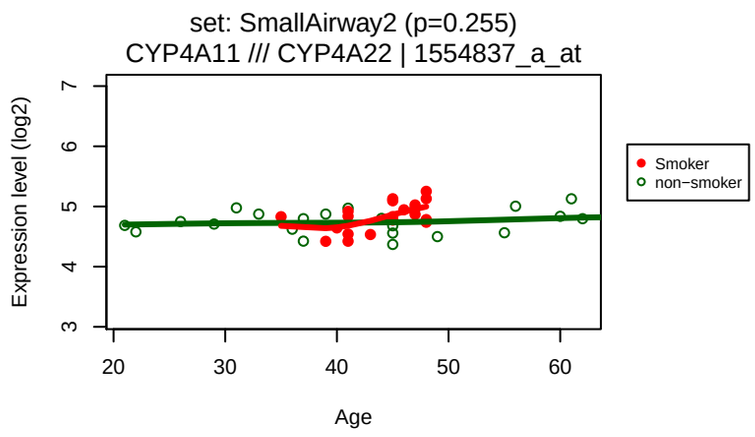
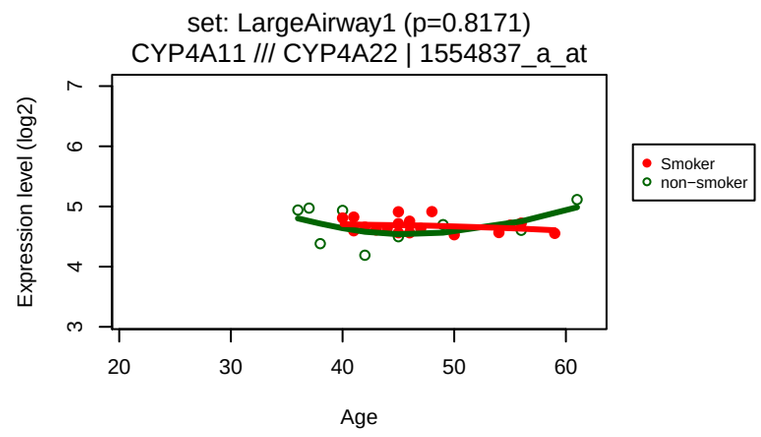

Gene expression of smokers and nonsmokers
gene: CYP4A11 /// CYP4A22 | 1554837_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2644
p=0.255
p=0.1722
p=0.8171
p=0.0338
p=0.6505
p=0.9211
p=0.7365
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

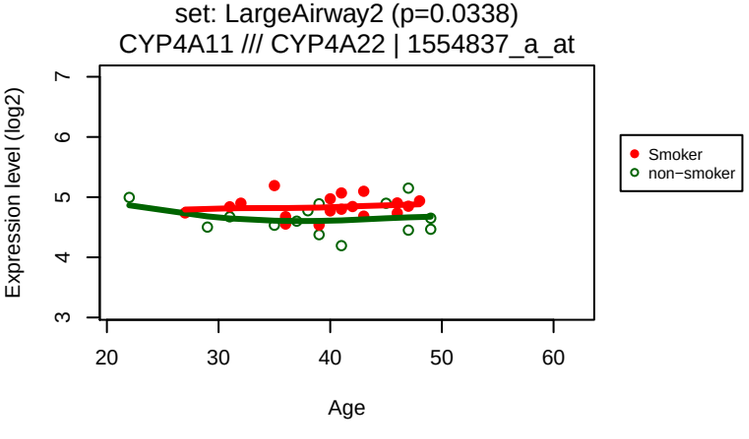

Gene expression ratio (smokers/nonsmokers)
gene: CYP4A11 /// CYP4A22 | 1554837_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2644
p=0.255
p=0.1722
p=0.8171
p=0.0338
p=0.6505
p=0.9211
p=0.7365
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

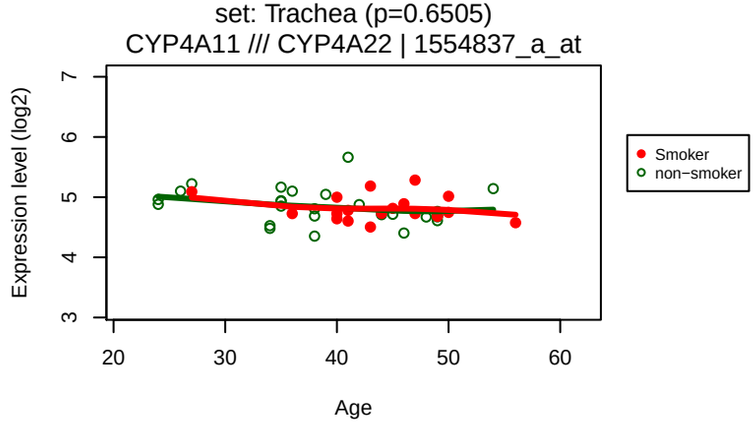

Gene expression ratio in Males
gene: CYP4A11 /// CYP4A22 | 1554837_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6696
p=0.1732
p=0.4223
p=0.22
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CYP4A11 /// CYP4A22 | 1554837_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3786
p=0.1364
p=0.0376
p=0.37
SmallAirway2
LargeAirway2
Trachea
Macrophage2

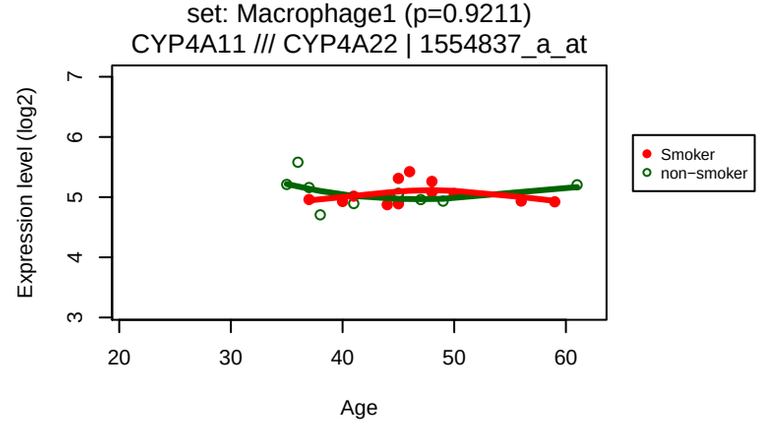

set: SmallAirway1 (p=0.2644)
CYP4A11 /// CYP4A22 | 1554837_a_at
Expression level (log2)
Age
Smoker
non−smoker

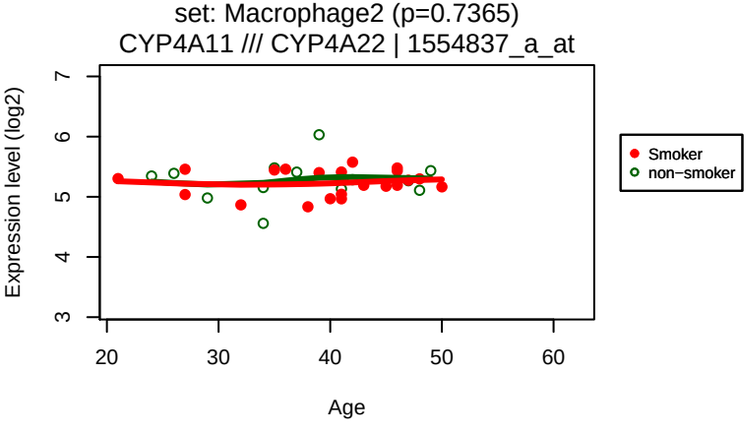

set: SmallAirway2 (p=0.255)
CYP4A11 /// CYP4A22 | 1554837_a_at
Expression level (log2)
Age
Smoker
non−smoker

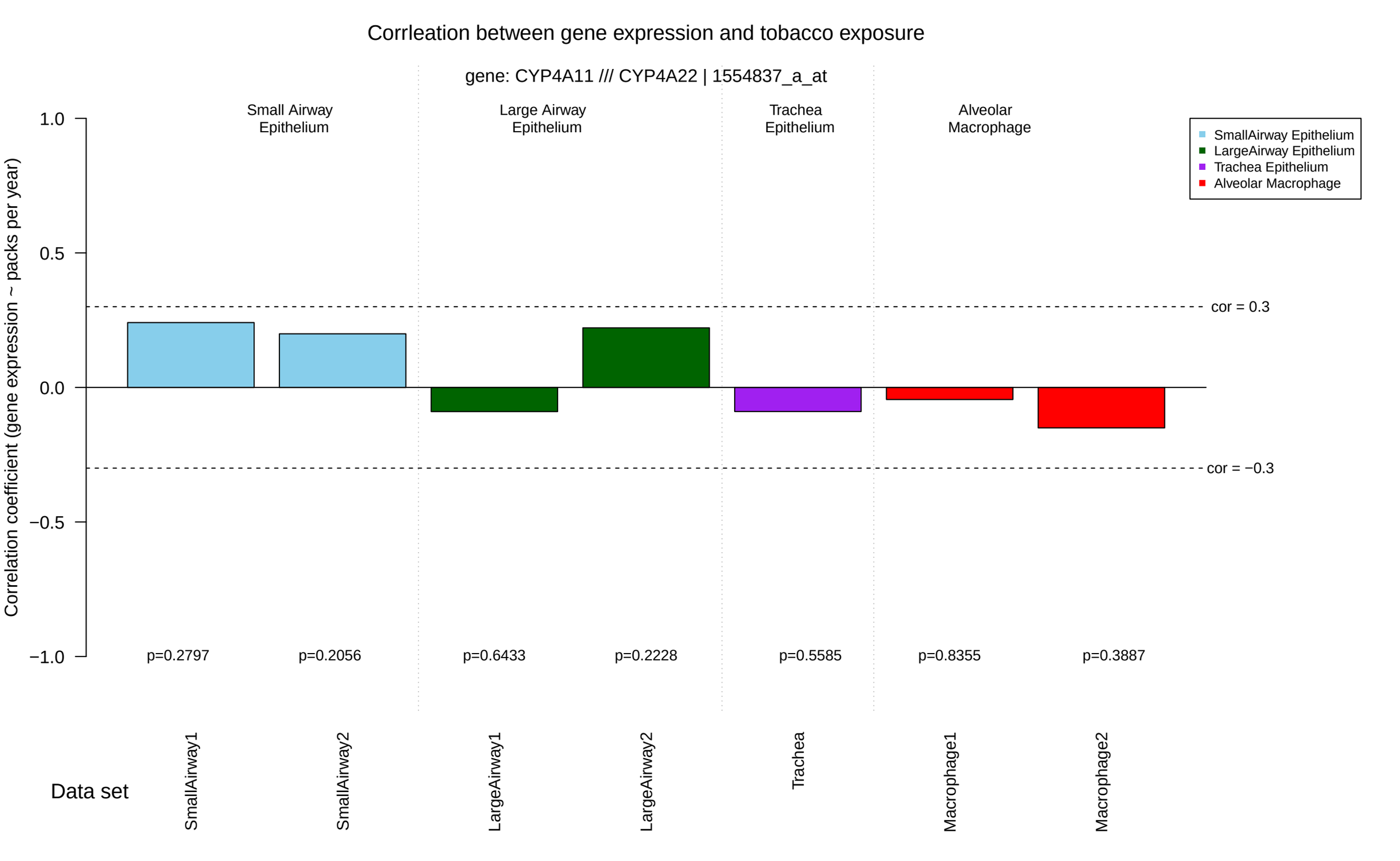

set: LargeAirway1 (p=0.8171)
CYP4A11 /// CYP4A22 | 1554837_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0338)
CYP4A11 /// CYP4A22 | 1554837_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6505)
CYP4A11 /// CYP4A22 | 1554837_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9211)
CYP4A11 /// CYP4A22 | 1554837_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7365)
CYP4A11 /// CYP4A22 | 1554837_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CYP4A11 /// CYP4A22 | 1554837_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2797
p=0.2056
p=0.6433
p=0.2228
p=0.5585
p=0.8355
p=0.3887
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2